

Institute of Actuaries of India

Subject CT3 – Probability & Mathematical Statistics

May 2012 Examinations Indicative Solutions

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other approaches leading to a valid answer and examiners have given credit for any alternative approach or interpretation which they consider to be reasonable

Soln.1 Data: 1,1,2,7,12,15,15,23,27,36,40,50,59

(a) Range: $59 - 1 = 58$

(b) Median = $\frac{(n+1)}{2}$ th value = $(14/2)$ th value = 7th value = 15

(c) Mean = 22.15385

[4]

Soln.2 Let μ_X and σ_X denote the mean and standard deviation for individual losses.

Let μ_N and σ_N denote the mean and standard deviation for number of claims.

We have been given:

$$\left. \begin{array}{l} \mu_X = 197,742 \\ \sigma_X = 52,414 \end{array} \right\} \& \left. \begin{array}{l} \mu_N = 7.6 \\ \sigma_N = 3.2 \end{array} \right\}$$

Let S denote the random variable for aggregate claims.

$$E(S) = \mu_X \cdot \mu_N \quad \text{[using formulas given in page 16 of the Tables]}$$

$$= (197,742)(7.6)$$

$$= 1,502,839.20$$

$$Var(S) = \mu_N \cdot \sigma_X^2 + \sigma_N^2 (\mu_X)^2 \quad \text{[using formulas given in page 16 of the Tables]}$$

$$= (7.6)(52,414)^2 + (3.2)^2 (197,742)^2$$

$$= 421,282,369,504.96$$

Hence, the mean and standard deviation of aggregate claims are 1,502,839.20 and 649,062.69 respectively.

[5]

Soln.3

(a) The regression model will be: $y_i = \alpha x_i + \epsilon_i$ for $i = 1, 2$ where ϵ_i are independent and normally distributed with mean 0 and variance 1.

(b) To minimize $L = \sum_{i=1}^2 (y_i - \alpha x_i)^2$ wrt α

$$\frac{dL}{d\alpha} = \sum_{i=1}^2 (-x_i)(y_i - \alpha x_i) = 0 \text{ implies } \hat{\alpha} = \frac{\sum x_i y_i}{\sum x_i^2}$$

$$\frac{d^2 L}{d\alpha^2} = \sum_{i=1}^2 x_i^2 > 0 \Rightarrow \text{minimum}$$

Therefore, the least square estimate of α is given as

$$\hat{\alpha} = \frac{\sum_{i=1}^2 x_i y_i}{\sum_{i=1}^2 x_i^2}$$

(c) The observed values of (x_i, y_i) are

$$(x_1, y_1) = (1, 0.6) \quad \& \quad (x_2, y_2) = (3, 1.8)$$

Therefore, the least square estimate of α is given as

$$\begin{aligned} \hat{\alpha} &= \frac{\sum_{i=1}^2 x_i y_i}{\sum_{i=1}^2 x_i^2} \\ &= \frac{(1)(0.6) + (3)(1.8)}{1 + 9} \\ &= \frac{6}{10} = 0.6 \end{aligned}$$

(d) We are given $y_3 = \alpha + \epsilon_3$

$$\text{Now, } \widehat{y}_3 = \hat{\alpha} + \hat{\epsilon}_3 = 0.6 + 0 = 0.6 \quad [\because \hat{\epsilon}_3 = E(\epsilon_3) = 0]$$

$$E(\widehat{y}_3) = E(\hat{\alpha} + 0) = \frac{\sum x_i E(y_i)}{\sum x_i^2} = \frac{\sum x_i (\alpha x_i + 0)}{\sum x_i^2} = \alpha$$

$$Var(\widehat{y}_3) = Var(\hat{\alpha}) = \frac{\sum x_i^2 \cdot Var(y_i)}{(\sum x_i^2)^2} \quad [Var(y_i) = Var(\epsilon_i) = 1]$$

$$= \frac{1}{\sum x_i^2} \quad \& \text{Cov}(y_1, y_2) = 0]$$

$$= \frac{1}{1+9} = 0.1$$

$$\text{We have: } y_3 - \widehat{y}_3 = (\alpha + \epsilon_3) - \hat{\alpha}$$

$$E(y_3 - \widehat{y}_3) = \alpha + E(\epsilon_3) - E(\hat{\alpha}) = \alpha + 0 - \alpha = 0$$

$$Var(y_3 - \widehat{y}_3) = Var(\epsilon_3) + Var(\hat{\alpha}) = 1 + 0.1 = 1.1 \quad [Cov(\epsilon_3, \hat{\alpha}) = 0]$$

So, $y_3 - \hat{y}_3 \sim N(0, 1.1)$ given $\epsilon_3 \sim N(0, 1)$ & $\hat{\alpha} \sim N(0, 0.1)$

Now, $P[|y_3 - \hat{y}_3| \leq z_{0.025} * \sqrt{1.1}] = 0.95$

Thus, a 95% prediction interval for y_3

$$\begin{aligned} & \hat{y}_3 \pm z_{0.025} \cdot \sqrt{1.1} \\ &= 0.6 \pm 1.96 * \sqrt{1.1} \\ &= 0.60 \pm 2.06 \\ &\Rightarrow (-1.46, 2.66) \end{aligned}$$

[11]

Soln.4 For a Normal random sample of size 100, a 90% confidence interval for the population mean is given by

$$\bar{x} \pm z_{0.05} \cdot \frac{s}{\sqrt{100}} = \bar{x} \pm 0.1645s$$

where $\bar{x}$ and s are the mean and standard deviation of the sample respectively.

We are given:

$$\left. \begin{aligned} \bar{x} - 0.1645s &= 20 \\ \bar{x} + 0.1645s &= 40 \end{aligned} \right\}$$

Solving the pair of equations we get:

$$\left. \begin{aligned} \bar{x} &= \frac{20 + 40}{2} = 30 \\ s &= \frac{40 - 20}{2 * 0.1645} = 60.79 \end{aligned} \right\}$$

Hence, a 95% confidence for the population mean will be given by:

$$\begin{aligned} &= \bar{x} \pm z_{0.025} \cdot \frac{s}{\sqrt{100}} \\ &= \bar{x} \pm 1.96 \cdot \frac{s}{10} \\ &= 30 \pm 0.196 * 60.79 \\ &= 30 \pm 11.91 \\ &\Rightarrow (18.09, 41.91) \end{aligned}$$

[5]

Soln.5. (a) As per the question, $X \stackrel{Y}{=} \begin{cases} 1 & \text{with probability} = P(\text{'Head' turns up}) = 0.5 \\ 0 & \text{with probability} = P(\text{'Tail' turns up}) = 0.5 \end{cases}$

i.e. in other words we can write $X = I.Y$ where I is an indicator random variable such that

$$P(I = 1) = P(\text{'Head' turns up}) = 0.5 \text{ [as coin is unbiased]}$$

$$\& \quad P(I = 0) = P(\text{'Tail' turns up}) = 0.5$$

$$\text{So, } M_X(t) = E(e^{tX})$$

$$= E[E(e^{tX} | I)]$$

$$= E(e^{tX} | I = 0) \cdot P(I = 0) + E(e^{tX} | I = 1) \cdot P(I = 1)$$

$$= 1 * 0.5 + M_Y(t) * 0.5$$

$$= 0.5 * [1 + M_Y(t)]$$

$$(b) \text{ Now } M_Y(t) = E(e^{tY})$$

$$= E[e^{t(\alpha + \beta N)}]$$

$$= e^{\alpha t} \cdot M_N(\beta t)$$

$$= e^{\alpha t} \cdot e^{\lambda(e^{\beta t} - 1)}$$

where $M_N(\cdot)$ is the mgf of random variable N

[using the formula given in page 7 of the tables]

$$\text{Then, } M_X(t) = 0.5[1 + M_Y(t)]$$

$$= 0.5[1 + e^{\alpha t} \cdot e^{\lambda(e^{\beta t} - 1)}]$$

$$= 0.5 + 0.5e^{-\lambda} \cdot e^{\alpha t} \exp[\lambda e^{\beta t}]$$

$$= 0.5 + 0.5e^{-\lambda} \sum_{j=0}^{\infty} e^{\alpha t} \cdot \frac{\lambda^j e^{j\beta t}}{j!}$$

$$= 0.5 + 0.5e^{-\lambda} \sum_{j=1}^{\infty} \frac{\lambda^{j-1} \exp[(\alpha + j - 1)\beta t]}{(j-1)!}$$

X takes values $0, \alpha, \alpha + \beta, \alpha + 2\beta, \alpha + 3\beta, \dots$

So, selecting coefficient of e^{xt} gives

$$P[X = 0] = 0.5$$

$$P[X = \alpha + \beta(x - 1)] = \frac{0.5e^{-\lambda}\lambda^{x-1}}{(x-1)!} \quad \text{For } x = 1, 2, 3, \dots$$

[Alternatively, the students can compute these probabilities directly using conditional probability:

X takes values $0, \alpha, \alpha + \beta, \alpha + 2\beta, \alpha + 3\beta, \dots$

$$P[X = 0] = P[X = 0 | \text{Coin turns up 'Tails'}] * P[\text{Coin turns up 'Tails'}] = 0.5$$

For $x = 1, 2, 3 \dots$

$$\begin{aligned}
 & P[X = \alpha + \beta(x - 1)] \\
 &= P[X = \alpha + \beta(x - 1) \mid \text{Coin turns up 'Heads'}] * P[\text{Coin turns up 'Heads'}] \\
 &= P[Y = \alpha + \beta(x - 1)] * P[\text{Coin turns up 'Heads'}] \\
 &= P[N = x - 1] * P[\text{Coin turns up 'Heads'}] \\
 &= \frac{e^{-\lambda} \lambda^{x-1}}{(x-1)!} * 0.5
 \end{aligned}$$

(c) X_1 takes the values 0, 1, 3, 5, 7.....

X_2 takes the values 0, 2, 5, 8, 11...

We can compute:

$$P(X_1 + X_2 \leq 5) = \sum_{c=0}^5 P(X_1 + X_2 = c)$$

x_1	$P(X_1 = x_1)$	x_2	$P(X_2 = x_2)$
0	0.50000	0	0.50000
1	0.18394	2	0.11157
3	0.18394	5	0.16735
5	0.09197	8	

Thus,

$X_1 + X_2$	(x_1, x_2)	Probability
0	(0, 0)	0.25000
1	(1, 0)	0.09197
2	(0, 2)	0.05578
3	(1, 2); (3, 0)	0.11249
4	not possible	0
5	(0, 5); (3, 2); (5, 0)	<u>0.15018</u>
		<u>0.66042</u>

$$\Rightarrow P(X_1 + X_2 \leq 5) = 0.66042$$

Hence,

$$\begin{aligned}
 P(X_1 + X_2 > 5) &= 1 - P(X_1 + X_2 \leq 5) \\
 &= 1 - 0.66042 = 0.33958
 \end{aligned}$$

[17]

Soln.6. (a) Let X be the quantum of error in marking an answer script. As per the question, the probability distribution of X is given as:

$$x: -5 \quad -4 \quad -3 \quad -2 \quad -1 \quad 0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5$$

$$p_x: \frac{1}{100} \quad \frac{1}{80} \quad \frac{1}{60} \quad \frac{1}{40} \quad \frac{1}{20} \quad p_0 \quad \frac{1}{20} \quad \frac{1}{40} \quad \frac{1}{60} \quad \frac{1}{80} \quad \frac{1}{100}$$

$$\text{Here: } p_0 = 1 - 2 * \left(\frac{1}{20} + \frac{1}{40} + \frac{1}{60} + \frac{1}{80} + \frac{1}{100} \right) = \frac{463}{600}$$

Therefore, the probability of no error will be $\frac{463}{600}$.

(b) We have $E(X) = 0$ as X is symmetric around 0.

$$\text{Var}(X) = E(X^2)$$

$$= \sum_{k=1}^5 2 \cdot k^2 \cdot \frac{1}{20k}$$

$$= \frac{1}{10} \sum_{k=1}^5 k$$

$$= 1.5$$

If S_{30} is the total error made in awarding marks over all 30 credit papers, the average quantum of error in a given student's final grade point average will be:

$$A_{30} = \frac{S_{30}}{30} = \frac{1}{30} \sum_{i=1}^{30} X_i$$

Applying the Central Limit Theorem, we can say $A_{30} \sim N\left(\mu, \frac{\sigma^2}{30}\right)$ where $\mu = E[X]$ and $\sigma^2 = \text{Var}[X]$.

Incorporating the computed values, we get $A_{30} \sim N(0, 0.05)$

(c) Now we want to compute:

$$P[-0.05 \leq A_{30} \leq +0.05]$$

$$= P\left[\frac{-0.05}{\sqrt{0.05}} \leq \frac{A_{30}}{\sqrt{0.05}} \leq \frac{0.05}{\sqrt{0.05}}\right]$$

[Using the results of part (b) we get $Z = \frac{A_{30}}{\sqrt{0.05}} \sim N(0,1)$]

$$\approx P[-\sqrt{0.05} \leq Z \leq \sqrt{0.05}]$$

$$= 2 * \Phi(0.224) - 1$$

$$= 0.177243.$$

This means that there is only a **17.7%** chance that a given student's final grade point average is accurate to within ± 0.05 .

[8]

Soln.7. (a) Let $X_1, X_2 \dots X_n$ are independent and identically distributed Poisson (λ) random variables.

$$P[X = x] = e^{-\lambda} \frac{\lambda^x}{x!} \text{ for } x = 0, 1, 2 \dots$$

The likelihood function is given by:

$$\begin{aligned} L(\lambda; x_1, x_2 \dots x_n) &= \prod_{i=1}^n P[X_i = x_i] \\ &= \prod_{i=1}^n e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} = e^{-n\lambda} \frac{\lambda^{\sum x_i}}{\prod x_i!} \end{aligned}$$

$$\text{So, } \log L(\lambda; \underline{x}) = -n\lambda + \log \lambda * \sum x_i - \log(\prod x_i!)$$

$$\text{Thus: } \frac{d \log L(\lambda; \underline{x})}{d\lambda} = 0 \Rightarrow -n + \frac{1}{\lambda} \sum x_i = 0$$

$$\Rightarrow \hat{\lambda} = \frac{1}{n} \sum x_i = \bar{x}$$

$$\text{Check: } \frac{d^2 \log L(\lambda; \underline{x})}{d\lambda^2} = -\frac{1}{\lambda^2} \sum x_i < 0 \Rightarrow \text{maximum}$$

Therefore, the maximum likelihood estimator of λ is $\hat{\lambda} = \bar{X}$.

(b) Suppose that rather than observing the random variables precisely, only the events " $X_i = 0$ " or " $X_i > 0$ " for $i = 1, 2 \dots n$ are observed.

$$\text{Define: } Y_i = I_{X_i > 0} = \begin{cases} 1 & \text{if } X_i > 0 \\ 0 & \text{if } X_i = 0 \end{cases}$$

$$\text{Then: } P[Y_i = 0] = P[X_i = 0] = e^{-\lambda}$$

$$P[Y_i = 1] = P[X_i > 0] = 1 - e^{-\lambda}.$$

Thus, the likelihood function in terms of $Y_1, Y_2 \dots Y_n$ would be:

$$L(\lambda; y_1, y_2 \dots y_n) = \prod_{i=1}^n P[Y_i = y_i]$$

$$= \prod_{i=1}^n (1 - e^{-\lambda})^{y_i} (e^{-\lambda})^{1-y_i}$$

$$= (1 - e^{-\lambda})^{\sum y_i} e^{-\lambda(n - \sum y_i)}$$

Taking logarithms,

$$\log L(\lambda; y_1, y_2 \dots y_n) = \left(\sum y_i \right) \cdot \log(1 - e^{-\lambda}) - \lambda \cdot \left(n - \sum y_i \right)$$

Thus: $\frac{d \log L(\lambda; \underline{y})}{d\lambda} = 0 \Rightarrow \hat{\lambda} = -\log(1 - \bar{y})$ where $\bar{y} = \frac{1}{n} \sum y_i$

Check: $\frac{d^2 \log L(\lambda; \underline{y})}{d\lambda^2} = -(\sum y_i) \frac{e^{-\lambda}}{(1 - e^{-\lambda})^2} < 0 \Rightarrow \text{maximum}$

Therefore, the maximum likelihood estimator of λ under the new observation scheme is

$$\hat{\lambda} = -\log(1 - \bar{Y}) = -\log\left(1 - \frac{1}{n} \sum I_{X_i > 0}\right) \equiv -\log\left(\frac{1}{n} \sum I_{X_i = 0}\right)$$

[10]

Soln.8. (a) The probability of the Type I error of Test 1 is $P_{\theta=0}[X_1 > 0.95] = 1 - \Phi(0.95)$

[Since under $\theta = 0$, $X_1 \sim N(0, 1)$]

The probability of the Type I error of Test 2 is $P_{\theta=0}[X_1 + X_2 > C] = 1 - \Phi\left(\frac{C}{\sqrt{2}}\right)$

[Since under $\theta = 0$, $X_1 + X_2 \sim N(0, 2)$]

We require: $1 - \Phi(0.95) = 1 - \Phi\left(\frac{C}{\sqrt{2}}\right)$

$$\Rightarrow C = 0.95 * \sqrt{2} = 1.3435$$

(b) The probability of the Type II error for Test 1 is given by:

$$\beta_1(\theta_1) = 1 - P_{\theta_1}[X_1 > 0.95]$$

$$= \Phi(0.95 - \theta_1) \text{ since } X_1 \sim N(\theta_1, 1)$$

The probability of the Type II error for Test 2 is given by:

$$\beta_2(\theta_1) = 1 - P_{\theta_1}[X_1 + X_2 > 1.3435]$$

$$\begin{aligned}
 &= \Phi\left(\frac{1.3435 - 2\theta_1}{\sqrt{2}}\right) \text{ since } X_1 + X_2 \sim N(2\theta_1, 2) \\
 &= \Phi(0.95 - \theta_1\sqrt{2})
 \end{aligned}$$

(c) $\Phi(x)$ is a monotonic increasing function of x .

Thus, for given $\theta_1 > 0$,

$$\begin{aligned}
 &\Phi(0.95 - \theta_1) > \Phi(0.95 - \theta_1\sqrt{2}) \\
 \Leftrightarrow &1 - \Phi(0.95 - \theta_1) < 1 - \Phi(0.95 - \theta_1\sqrt{2})
 \end{aligned}$$

This means $1 - \beta_1(\theta_1) < 1 - \beta_2(\theta_1)$

In fact it can be generalised that for any given $\theta > 0$,

$$1 - \beta_1(\theta) < 1 - \beta_2(\theta)$$

Therefore, Test 2 is more powerful test than Test 1.

[6]

Soln.9. We are testing the following Null Hypothesis against Alternate Hypothesis:

H_0 : there is no association between factory and defectiveness in the product (i.e. they are independent)

H_1 : there is an association between factory and defectiveness in the product (i.e. they are not independent)

The expected values can be calculated as follows: (Row Total) * (Column Total) / Grand Total and thus we get:

	Factory A	Factory B
Non-defective	1,901	1,901
Defective	99	99

The degrees of freedom are $(2 - 1) * (2 - 1) = 1$

$$\begin{aligned}
 \chi^2 &= \sum_1^4 \left[\frac{(O-E)^2}{E} \right] \\
 &= \frac{(1816-1901)^2}{1901} + \frac{(1986-1901)^2}{1901} + \frac{(184-99)^2}{99} + \frac{(14-99)^2}{99}
 \end{aligned}$$

$$= 3.80063 + 3.80063 + 72.97980 + 72.97980$$

$$= 153.56$$

Decision: Since this exceeds even the 0.05% critical value of 12.12, we have very strong evidence even at the 0.05% level to reject the null hypothesis H_0 . We therefore conclude that there is an association between factory and defectiveness in the product.

[5]

Soln.10. The 95% confidence interval for β is given by:

$$\hat{\beta} \pm t_{0.025, 5} * se(\hat{\beta}) \text{ i.e. } \hat{\beta} \pm 2.57 * se(\hat{\beta}) \quad [\because t_{0.025, 5} = 2.57]$$

$$\text{We are given: } \hat{\beta} - 2.57 * se(\hat{\beta}) = 0.030 \quad \& \quad \hat{\beta} + 2.57 * se(\hat{\beta}) = 0.088$$

$$\text{Thus: } \hat{\beta} = 0.059 \quad \& \quad se(\hat{\beta}) = \frac{0.088 - 0.030}{2.57 * 2} = 0.011284.$$

$$\text{We can write } se(\hat{\beta}) = \frac{\hat{\sigma}}{\sqrt{S_{xx}}} \text{ where } S_{xx} = \sum (x_i - \bar{x})^2$$

$$\text{This means: } \hat{\sigma} = se(\hat{\beta}) * \sqrt{S_{xx}} = 0.011284 * \sqrt{280000} = 5.97093$$

$$\text{Thus, } \hat{\sigma}^2 = 35.652.$$

$$\text{Again, we can write } \hat{\sigma}^2 = \frac{1}{7-2} \left[S_{yy} - \frac{S_{xy}^2}{S_{xx}} \right]$$

$$\text{Thus, } S_{yy} = 5 * 35.652 + \frac{16500 * 16500}{280000}$$

$$= 1150.581.$$

Therefore, the portion of the total variability of the responses explained by the model

$$R^2 = \frac{S_{xy}^2}{S_{xx} * S_{yy}}$$

$$= \frac{16500 * 16500}{280000 * 1150.581}$$

$$= 0.845 \text{ (i.e., 84.5\%)}$$

[5]

Soln.11. (a) Under Design 1, the weights relate to different sets of individuals. As no two individuals were the same, we can consider the two samples to be independent.

Under Design 2, the weights relate to same sets of individuals measured once before the campaign and then at the end of the campaign. Consequently we can't consider the two samples to be independent.

(b) **Design 1**

Here under Design 1, the two samples are independent. So, we will be performing two-sample analysis. Using the given summary statistics, we get:

$$\bar{X}_1 = \frac{870}{5} = 174$$

$$\bar{X}_2 = \frac{850}{5} = 170$$

$$\sum (X_1 - \bar{X}_1)^2 = \sum X_1^2 - n \cdot \bar{X}_1^2 = 152,324 - 5 * 174 * 174 = 944$$

$$\sum (X_2 - \bar{X}_2)^2 = \sum X_2^2 - n \cdot \bar{X}_2^2 = 145,366 - 5 * 170 * 170 = 866$$

The 95% confidence interval for the mean weight loss during the campaign is given by:

$$\mu_1 - \mu_2: (\bar{X}_1 - \bar{X}_2) \pm t_{0.025;df} * s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Here: $df = (n_1 - 1) + (n_2 - 1) = 4 + 4 = 8$; $t_{0.025;df} = 2.306$;

$$s_p^2 = \frac{\sum (X_1 - \bar{X}_1)^2 + \sum (X_2 - \bar{X}_2)^2}{(n_1 - 1) + (n_2 - 1)} = \frac{944 + 866}{4 + 4} = 226.25$$

$$\text{i.e. } \mu_1 - \mu_2: (174 - 170) \pm 2.306 * \sqrt{226.25} * \sqrt{\frac{1}{5} + \frac{1}{5}} = 4 \pm 21.94$$

$$\Rightarrow (-17.94, 25.94)$$

Design 2

Here under Design 2, the two samples are not independent. This is an example of paired data and is analyzed using the differences $D = X_1 - X_3$.

The 95% confidence interval for the mean weight during the campaign under Design 2:

$$\Delta: \bar{D} \pm t_{0.025;df} * \frac{S_D}{\sqrt{n}}$$

where $n = 5$; $df = 5 - 1 = 4$; $S_D^2 = \frac{1}{n-1} \sum (D - \bar{D})^2$

Using the given summary statistics, we get:

$$\bar{X}_1 = \frac{870}{5} = 174$$

$$\bar{X}_3 = \frac{850}{5} = 170$$

$$\bar{D} = \bar{X}_1 - \bar{X}_3 = 174 - 170 = 4$$

$$\begin{aligned} \sum D^2 &= \sum (X_1 - X_3)^2 = \sum X_1^2 - 2 \cdot \sum X_1 X_3 + \sum X_3^2 \\ &= 152,324 - 2 * 149,032 + 145,878 = 138 \end{aligned}$$

$$\sum (D - \bar{D})^2 = \sum D^2 - n \cdot \bar{D}^2 = 138 - 5 * 4 * 4 = 58$$

$$S_D^2 = \frac{1}{5-1} \sum (D - \bar{D})^2 = 14.5$$

Thus, the 95% confidence interval for the mean weight will be:

$$\Delta: 4 \pm 2.776 * \sqrt{\frac{14.5}{5}} = 4 \pm 4.73 \Rightarrow (-0.73, 8.73)$$

[Alternatively, the students can also compute $\bar{D}$ and $\sum (D - \bar{D})^2$ directly from raw data]

- (c) For both cases, we observe the interval contains zero. So, we can conclude that there is no evidence at the 5% level to suggest a significant difference in the mean weights level pre- and post- campaign. This means that the fitness campaign might not have produced its desired results.

Further we note that the confidence interval obtained using results obtained under Design 1 is much wider than that obtained using results obtained under Design 2. This is as expected. Under Design 1, we are measuring weights of completely different set of individuals and comparing the impact. There will be a component of sampling error here which will make the confidence interval wider. Such sampling error will not exist under sampling Design 2 as we are working with the same individuals.

[11]

Soln.12. (a) We are testing the hypotheses:

$H_0: \mu_1 = \mu_2 = \mu_3$ i.e. there is no significant difference between the mean number of items produced for all days under each music played

Here:

- μ_1 , the mean number of items produced for all days when country music is played;
- μ_2 , the mean number of items produced for all days when rock music is played; and
- μ_3 , the mean number of items produced for all days when classical music is played.

H_1 : There is a difference between the mean number of items produced for all days under each music played

Test Statistic: $F = \frac{\text{variability among the sample means}}{\text{variability due to chance}}$

$$SS(T) = 8,042,637 - \frac{(9,813)^2}{12} = 18056.25$$

$$SS(B) = \frac{(3295)^2}{4} + \frac{(3101)^2}{4} + \frac{(3417)^2}{4} - \frac{(9813)^2}{12} = 12,698$$

$$SS(R) = SS(T) - SS(B) = 18056.25 - 12698 = 5358.25$$

ANOVA TABLE

Sources of Variation	DF	SS	MSS	F
Between Music	2	12,698.00	6,349.0000	10.66412
Residual	9	5,358.25	595.3611	
Total	11	18,056.25		

Calculations from $F = 10.66$, Critical Value = 4.256. As the Critical Value is less than calculated value we should reject H_0 .

Conclusion: At the 5% level of significance we can conclude that at least two of the means are not equal.

(b) μ_1 = Country Mean, μ_2 = Rock Mean, μ_3 = Classical Mean

Now examine each of the pairs in order to see whether the means are the same or not using a one-sided test.

Testing Rock Music is significantly worse than other two:

Case 1:

$$H_0: \mu_1 = \mu_2 \quad H_1: \mu_2 < \mu_1$$

$$\sigma^2 = \frac{SS(R)}{n-k} = \frac{5358.25}{12-3} = 595.3611$$

$$\text{Test Statistic} = \frac{(823.75 - 775.25) - 0}{\sqrt{595.3611 * 0.5}} = 2.81$$

Critical Value $t_9 = 1.833$ at 5% level

Since the critical value is less than statistical value which is 2.81 therefore we have sufficient evidence to reject H_0 . Hence we get the result as $\mu_2 < \mu_1$.

Case 2:

$$H_0: \mu_2 = \mu_3 \quad H_1: \mu_2 < \mu_3$$

$$\text{Test Statistic} = \frac{(775.25 - 854.25) - 0}{\sqrt{595.3611 * 0.5}} = -4.5788$$

Critical Value $t_9 = 1.833$ at 5% level

Since the critical value is less than statistical value which is 4.5788, therefore we have sufficient evidence to reject H_0 . Hence we get the result as $\mu_2 < \mu_3$.

Conclusion: From the above analyses we see that rock music is the worst in terms of worker productivity.

(c) Testing for best music

We will test this between Country and Classical music.

$$H_0: \mu_1 = \mu_3 \quad H_1: \mu_1 < \mu_3$$

$$\text{Test Statistic} = \frac{(823.75 - 854.25) - 0}{\sqrt{595.3611 * 0.5}} = -1.7677$$

Critical Value $t_9 = 1.833$ at 5% level

Since the critical value is greater than statistical value which is 1.7677, therefore we have insufficient evidence to reject H_0 . Hence we get the result as $\mu_1 = \mu_3$.

Conclusion: Classical music may be the best, but perhaps country music could also be. So statistically it is difficult to tell which music is the best.

[Alternately, the students can also perform a two tailed test for testing difference between μ_1 and μ_3]

[12]

Total Marks - 100]

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